

# Galois Theory of the Pandrosion Quotient Tower

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## Abstract

The Galois group of a polynomial  $P$  acts naturally on the Pandrosion quotients via  $\sigma : Q(\alpha_j, z) \mapsto Q(\alpha_{\sigma(j)}, z)$ . The Pandrosion field  $F_P = P'/P$ , the Turán SOS  $\sum Q_j^2$ , and the discriminant  $\Delta = \prod(\alpha_i - \alpha_j)^2$  are all Galois-invariant—they are rational even when the individual roots are not. The cubic resolvent of a quartic arises as the orbit polynomial of the pairing partition under  $S_4$ . We show that  $\sqrt{\Delta}$  is rational if and only if  $\text{Gal}(P) \subseteq A_n$ , linking the Pandrosion cross-Wronskian (Paper 64) to the alternating group.

## 1 The Galois action on quotients

**Definition 1.1.** For  $P(z) = \prod(z - \alpha_j) \in \mathbb{Q}[z]$  irreducible, the Galois group  $G = \text{Gal}(P)$  acts on the roots:  $\sigma(\alpha_j) = \alpha_{\sigma(j)}$ . This extends to the Pandrosion quotients:

$$\sigma : Q(\alpha_j, z) = \frac{P(z)}{z - \alpha_j} \mapsto Q(\alpha_{\sigma(j)}, z) = \frac{P(z)}{z - \alpha_{\sigma(j)}}. \quad (1)$$

## 2 Galois-invariant Pandrosion expressions

**Theorem 2.1** (Rationality). *The following expressions are  $G$ -invariant and hence lie in  $\mathbb{Q}(z)$ :*

1.  $P'(z) = \sum_{j=1}^n Q(\alpha_j, z)$  (orbit sum);
2.  $T(z) = P'(z)^2 - P(z)P''(z) = \sum_{j=1}^n Q(\alpha_j, z)^2$  (orbit norm / Turán);
3.  $\text{Res}(P, Q) = \prod_j Q(\alpha_j, z_0)$  for any  $Q \in \mathbb{Q}[z]$  (orbit product / resultant);
4.  $\Delta(P) = \prod_{i < j} (\alpha_i - \alpha_j)^2$  (discriminant).

*Proof.* Each expression is a symmetric function of the roots (or the full alternating product squared), hence fixed by  $G$ .  $\square$

## 3 The alternating criterion

**Theorem 3.1.**

$$\text{Gal}(P) \subseteq A_n \iff \sqrt{\Delta(P)} \in \mathbb{Q}. \quad (2)$$

*Proof.*  $\sqrt{\Delta} = \prod_{i < j} (\alpha_i - \alpha_j)$  changes sign under odd permutations. It is fixed by  $G$  iff every element of  $G$  is even, i.e.,  $G \subseteq A_n$ . From Paper 64:  $W(Q_i, Q_j) = (\alpha_i - \alpha_j)R_{ij}^2$ , so  $\sqrt{\Delta}$  is a product of cross-Wronskian factors.  $\square$

Verified:  $\Delta(z^3 - 2) = -108$  (not a square:  $\text{Gal} = S_3 \not\subseteq A_3$ );  $\Delta(z^4 + 1) = 256 = 16^2$  (square:  $\text{Gal} = V_4 \subseteq A_4$ );  $\Delta(z^3 - 3z + 1) = 81 = 9^2$  (square:  $\text{Gal} = C_3 \subseteq A_3$ ). Criterion verified 7/7 on test polynomials, including the subtle  $n = 2$  case ( $S_2 \not\subseteq A_2$  since  $A_2 = \{\text{id}\}$ ).

## 4 Resolvents as orbit polynomials

**Proposition 4.1** (Cubic resolvent). *For a quartic  $P(z) = z^4 + bz^2 + c$  with roots  $\alpha_1, \dots, \alpha_4$ , the cubic resolvent is the orbit polynomial:*

$$R(t) = \prod_{k=1}^3 (t - \theta_k), \quad \theta_k = (\alpha_{i_k} + \alpha_{j_k})(\alpha_{l_k} + \alpha_{m_k}), \quad (3)$$

where the three pairings  $\{ij|lm\}$  are the orbits of  $S_4/V_4$ .

| $P(z)$           | Resolvent $R(t)$  | Notes                          |
|------------------|-------------------|--------------------------------|
| $z^4 - 5z^2 + 6$ | $t^3 + 10t^2 + t$ | $\theta = 0, -5 \pm 2\sqrt{6}$ |
| $z^4 - z^2 - 1$  | $t^3 + 2t^2 + 5t$ | $\theta = 0, -1 \pm 2i$        |
| $z^4 - 2$        | $t^3 + 8t$        | $\theta = 0, \pm 2\sqrt{2}i$   |

## 5 Orbit structure by group type

| Group | $Q$ -orbit structure        | Pandrosion invariants          |
|-------|-----------------------------|--------------------------------|
| $S_n$ | one orbit of size $n$       | $\sum Q_j, \sum Q_j^2, \Delta$ |
| $A_n$ | one orbit + $\sqrt{\Delta}$ | above $+\sqrt{\Delta}$         |
| $C_n$ | DFT diagonalizes            | $\sum \omega^{jk} Q_j$         |
| $D_n$ | reflection orbits           | cosine sums of $Q$ -values     |

## References

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